

Rendering Contention Channel Made Practical in Web Browsers

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Rendering Contention Channel Made Practical in Web Browsers

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Browser rendering utilizes hardware resources shared within and across browsers to display web contents, thus inevitably being vulnerable to side channel attacks. Prior works have studied rendering side channels that are caused by rendering time differences of one frame, such as URL color change. However, it still remains unclear how rendering contentions play a role in side-channel attacks and covert communications.

In this paper, we design a novel rendering contention channel. Specifically, we stress the browser's rendering resource with stable, self-adjustable pressure and measure the time taken to render a sequence of frames. The measured time sequence is further used to infer any co-rendering event of the browser.

To better understand the channel, we study its cause via a method called single variable testing. That is, we keep all variables the same but only change one to test whether the changed variable contributes to the contention. Our results show that CPU, GPU and screen buffer are all part of the contention.

To demonstrate the channel's feasibility, we design and implement a prototype, open-source framework, called *SIDER*, to launch four attacks using the rendering contention channel, which are (i) cross-browser, cross-mode cookie synchronization, (ii) history sniffing, (iii) website

fingerprinting, and (iv) keystroke logging. Our evaluation shows the effectiveness and feasibility of all four attacks.

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